SIH 2026 · PROBLEM STATEMENT 26147 · SPONSOR NTRO · THEME SPACE TECHNOLOGY · CATEGORY SOFTWARE

Project SANJAYA — explained end to end

What we are doing · how we are doing it · the complete user journey · every feature and USP with its tech stack · and a strict, rubric-based comparison against all 44 distinct public repositories found for this PS.

Everything in this document is sourced from the project's own evidence — PRD v2.4, the engine/UI build state (27 Sep 2026), the measured feasibility results (PRD §16) — and from a computed scoring of the 52-entry public-repository snapshot dated 26 Sep 2026 (_build/data.json), deduplicated to 44 distinct projects. Competitor statements are README-and-metadata claims, not audited code; the rubric, weights and hygiene rules are printed so the rating can be reproduced or argued with.

Part	Answers
A · What we are doing	The problem in plain words, what SANJAYA is, and what we deliberately do not do.
B · How we are doing it	The method (propose → verify → prove → sign → re-check), the engine loop, and the 10-week build flow.
C · The user journey, A to Z	Every step from opening the app to handing a receiver to a sensor — what the user does, what happens inside, with times.
D · Feature → tech stack	Each capability and USP mapped to the exact technology that implements it, with honest build status.
E · Comparison with all public repos	52 entries → 44 scored on a published rubric; field aggregates; leaderboard; head-to-head matrix; strict per-repo verdicts on who is better and why.

PART A

What we are doing

The problem, the vocabulary, and the product in plain words

The assignment (PS 26147, NTRO). Build a GUI-based tool that takes a recording of an unknown radio transmission — a **.IQ** file (raw samples of the radio wave, stored as number pairs) or a **.wav** file — and automatically works out how the transmitter was built: the **sampling frequency** it was recorded at, the **modulation** (how bits ride on the wave: PSK, QAM, FSK), the **error-correction code** (FEC: convolutional/Viterbi, Reed-Solomon, concatenated, LDPC), the **interleaving** (how bits are shuffled to survive bursts of noise: block, convolutional, diagonal, pseudo-random), and the **bit-stream structure** — where the header ends and the payload begins. Nothing is known in advance: no metadata, no manual, any sensor, anywhere.

Why that is genuinely hard. (1) **Everything is blind** — even the sampling frequency may be unknown, so every parameter must be inferred from the samples themselves. (2) **Wrong answers can look perfect** — a wrongly-guessed error-correction code can "correct" its own errors and produce clean output; a wrongly-guessed scrambler is invisible in the waveform. (3) **The problem statement itself warns** that data is "often insufficient for fine-grain analysis", arrives "from different sensors and different locations", and this "reduces the accuracy and confidence of interpretation".

What SANJAYA is. An offline, air-gap-ready application that turns such a recording into a complete, **proven** transmitter recipe — and proves it by an unusual route: it **re-builds the transmitter** from its own answer, re-generates the wave from the decoded bits, subtracts it from the recording, and shows that what remains is pure noise, layer by layer. The result ships as a signed **Decode Certificate** that anyone can re-check in seconds without trusting us. When the recording is too short or too noisy to decide, it says so honestly and computes **how much more data** would settle the question. It can also hand the recovered recipe to a designated sensor as a ready-to-run receiver.

> **One sentence.** Other tools give an answer; SANJAYA gives an answer, the evidence for every parameter, a mathematical demonstration that the answer explains the recording, a signature over that demonstration, and an honest "undecided" when the data does not support one.

What we deliberately do not do (scope discipline, stated up front): no payload decryption (structure only), no cloud/accounts/telemetry, no always-on network, no speech content or speaker identification, no OFDM unless the sponsor asks (planned as a stretch), no live SDR control as a requirement.

PART B

How we are doing it

The method, the engine loop, and the build flow

The core idea: finding is hard, checking is cheap. Discovering a transmitter recipe may take minutes of search over a huge hypothesis space. But **checking** a candidate recipe is fast and mathematical: re-synthesise the waveform from the decoded bits and see whether it explains the recording. SANJAYA separates the two: fast, fallible searchers **propose**; physics and algebra **verify**; only a verified, layer-proven recipe is ever reported. A model error costs time — never correctness.

Stage of the loop	What happens	Why it exists
1 · Observe + measure	Waterfall/PSD; detect emitters; estimate Fs, symbol rate, CFO, SNR, roll-off — each with an uncertainty and a 95 % interval	Every later decision needs calibrated inputs
2 · Hypothesise	Best-first tree search over modulation × ambiguity × interleaver × code × scrambler × framing, with cheap calibrated pruning between stages	The space is astronomically large; search must be ordered and prunable
3 · Demodulate + unwrap	Sync, equalise, demodulate to soft bits; identify the interleaver jointly with the code; recover the scrambler; peel concatenated codes	The PS chain (ii)–(iv); joint ID is what makes it work with little data
4 · Structure	Frames, sync word, counters, length fields; CRC polynomial solved in GF(2); HEADER / PAYLOAD labels	PS clause (v) — and it feeds the proof
5 · Prove (layered)	Re-synthesise from the decoded bits, subtract, test the residual: PHY at the measured noise floor; coding by parity-membership; structure by sync/CRC/counters	The part nobody else has: each layer gets a test that has power there
6 · Sign + re-check	Ed25519-signed Decode Certificate; anyone re-runs the proof from the file alone in seconds	Turns “trust me” into “check me”
7 · Learn (stretch)	Proof-gated self-labelling improves the models on real data with a bounded label-error rate	The system improves in use, without human labels

The layered proof, precisely. At high SNR the waveform alone cannot distinguish a correct recipe from a wrong-but-plausible one — measured: at 12 dB the residual of the *wrong* “uncoded” hypothesis sits 0.001 dB from the noise floor, indistinguishable from the right recipe’s 0.002 dB. So the proof is **layered**: PHY certified by waveform tests (T1–T4), coding by parity tests (K1–K3: unsatisfied-parity fraction 0.0003 for the true code vs 0.494 for random bits — about 88 standard deviations of separation), structure by sync recurrence, CRC pass fractions (20/20 CRC-16 passes by chance is 2^{-320}) and counters. Because the search tries many hypotheses, the certificate records **K**, the number of proof attempts, and carries a false-PASS bound of ≤ 10^{-4} per certificate (per-attempt bound 1.5×10^{-7} from a planned 2,000,000-attempt near-miss campaign; ≤8 attempts fast path, ≤64 deep).

The build flow (10 weeks, gate-driven). Six members × 25 h/week × 10 weeks = 37.5 person-weeks of capacity. The committed **threshold** configuration is 31.65 pw at **73 % schedule confidence** (200,000-run Monte Carlo, NASA-JCL style); the fuller 34.35 pw baseline is the goal. Gates: **G0** 4 Oct baseline → **G1** 11 Oct vertical slice → **G2** 25 Oct blind coding + first Meteor-M satellite capture → **G3** 8 Nov Reed-Solomon + Meteor fully blind → **G4** 22 Nov LDPC, installers, all requirements pass → **G5** 29 Nov independent KPI report + false-PASS campaign. A pre-agreed 8-rung descoping ladder (2.70 pw) protects the committed scope; PS coverage and the proof are never cut. **Already ahead of plan:** the G1 vertical slice and the G2 blind search are working now (Part C).

PART C

The user journey, A to Z

From the moment SANJAYA opens to the moment a sensor runs a receiver

#	User does / sees	What happens inside	Time
0	Opens SANJAYA (desktop app or browser)	Engine sidecar starts, binds to 127.0.0.1 only, token + Host/Origin checks; zero outbound network packets by design (air-gap ready); UI shows the frozen 5-step flow	Ready ≤ 5 s
1	Drops a file: .wav, .iq or SigMF	Format forensics: WAV variants (stereo-IQ, 8–32-bit, float, RF64/BW64 over 4 GB, HDSDR auxi chunk, KiwiSDR GPS chunk); raw-IQ dtype sniffing across 7 dtypes × endianness × I/Q order, each with a confidence; files to 8 GB memory-mapped; waterfall + PSD render immediately	100 MB interactive < 2 s (tiles measured 0.62 s)
2	Sees the triage verdict	Digital / voice / CW / noise gate before any decoding; abstains when the file is too short instead of hallucinating a decode	< 1 s
3	Sees the Scene view	OS-CFAR detection finds every emitter, polyphase channelisation splits them, bursts are segmented — a recording with 4 signals becomes 4 per-emitter jobs	Seconds
4	Reads the measurement card	Sampling frequency (header, or back-solved from known standards — target ≤ 10 ppm), symbol rate (family-specific estimators; P95 relative error ≤ 10^{-4}) — every value ships with uncertainty u and a 95 % interval	Seconds
5	Reads the modulation verdict + candidates	Three experts fused: feature GBDT, compact CNN (< 3 M params, ONNX), and a physics likelihood that cannot be fooled by the sim-to-real gap; output is a calibrated probability plus a conformal set; unknown signals flagged NOVEL by an empty set	Fast-path allocation 1.5 s
6	Watches demodulation proceed	Feedforward block synchronisers + C++ tracking loops, CMA → DD-LMS equaliser, OQPSK offset handling, FSK tone bank, burst acquisition; emits soft bits plus the explicit ambiguity set (phase rotations × conjugation) instead of one guess	≈ 1 s per 10^6 samples (allocation)
7	Sees the recipe assemble	Interleaver identified jointly with the code by soft-rank methods (needs ≤ 3N rows vs 44,411 rows for hard rank at period 1,024); convolutional K 3–9, Reed-Solomon basis/depth, LDPC catalogue by syndrome; scrambler position and polynomial recovered incl. the post-FEC coset case; concatenated codes peeled inner-first	Blind search measured 2–13 s per file (current engine)
8	Sees HEADER / PAYLOAD labels	Frame detection, sync-word discovery, counters and length fields, CRC polynomial solved in GF(2); bit view with entropy and autocorrelation	Sub-second
9	Presses Prove (or Autopilot does)	The decisive step: re-encodes the decoded bits through the recovered chain, re-generates the waveform, applies the estimated channel, subtracts → residual must sit at the measured noise floor (PHY T1–T4), satisfy the code's parity checks (K1–K3) and the structure checks (S1–S4). Near-miss hypotheses get the identical model, so no extra freedom favours the winner. Failed branches loop back into the search	Fast path target P95 < 10 s per 10^6 samples
10	Receives the Decode Certificate	Canonical JSON signed with Ed25519 (key in the OS key store): input SHA-256, engine version + config hash, full recipe, per-layer verdicts, every statistic with its threshold, the near-miss battery with rejected/not-rejected, K attempts and the ≤ 10^{-4} false-PASS bound; analyst overrides marked	Instant
11	Forwards the certificate; a colleague re-checks it	**sanjaya verify file cert.json** re-runs the synthesis and every test from the recording alone — no search, no trust, no separate tooling	Measured ≈ 1.7 s
12	Exports for the sensors	Recipe JSON (SigMF-compatible), bits, header/payload CSV, SigMF annotations, certificate PDF — and the Recipe Compiler emits a self-contained GNU Radio 3.10 flowgraph (.grc, stock blocks + embedded Python) that must reproduce the bits bit-exactly at the release gate before it ships	Seconds
13	File was too short? Reads the honest answer	"Undecided" with a Data-Sufficiency note: the evidence ledger predicts how much more signal is needed (P50/P90 duration) and issues a tasking note — centre frequency, occupied bandwidth, minimum sample rate (≥ 1.25× bandwidth), minimum SNR, and the best sensor to use	—
14	Has several short captures from different sites?	Stitching: associate captures, normalise each sensor's clock/noise via the Sensor Twin, align, pool log-likelihoods; the per-capture proof ejects wrong associations. Validation target: one transmission recorded by ≥ 3 public KiwiSDR receivers in different countries, GPS-timestamped	—

The honest state of this journey today (27 Sep 2026): steps 0–2 and 10–12 are implemented and verified end to end in Chrome (drop → blind decode → proof residual at the noise floor → signed certificate, re-verified in ≈ 1.7 s) for the QPSK engine slice; the blind search covers symbol rate, roll-off, CFO, clock ppm, frame + sync, block interleaver, rate-1/2 convolutional codes (K 3–9), frame-synchronous scramblers, phase ambiguity, counters and CRC. OQPSK/QAM/FSK/MSK, Reed-Solomon, LDPC, sufficiency and stitching are gated at G2–G4. Every limit is written down — that is deliberate.

PART D

Every feature and USP → its tech stack

With honest build status as of 27 Sep 2026

Feature / USP	What it does for the user	Tech stack (what actually implements it)	Status
Ingest + forensics (R-01)	Opens any .wav / raw .iq / SigMF; sniffs dtype, endianness, I/Q order; survives hostile files	Python 3.10, NumPy memory-maps, stdlib wave/struct; SigMF 1.x parser; range-checked headers	Working (browser + engine)
Triage gate (R-15)	Digital vs voice vs CW vs noise before decoding; abstains when unsure	Gradient-boosted trees → ONNX Runtime; rule-based fallback ported from ui/src/lib/triage.ts	Heuristic working; model G3
Scene / detection (R-17)	Finds and separates multiple emitters and bursts in one file	NumPy/SciPy: OS-CFAR on PSD/spectrogram, polyphase down-conversion, burst segmentation	Single-emitter now; multi G3
Measurement (R-02/03)	Fs, symbol rate, CFO, SNR, roll-off — each with uncertainty + 95 % interval	NumPy/SciPy: Welch PSD, M2M4 SNR, cyclic lines, family-specific rate estimators, jackknife u, Cramér–Rao cross-check	QPSK class working; families G2–G3
Modulation ID (R-04)	10 classes (BPSK–64QAM, FSK, MSK) with calibrated confidence + conformal set	E1 GBDT (scikit-learn → ONNX) + E2 1-D ResNet / 2-D CNN < 3 M params (PyTorch train → ONNX Runtime) + E3 physics likelihood (NumPy); temperature scaling + logistic stacking	E1+E3 v0 at G1; fusion G3
Demodulation (R-07)	PSK/OQPSK/QAM/FSK/MSK receivers incl. bursts; soft LLRs + ambiguity set	NumPy/SciPy feedforward sync (Oerder–Meyr, Viterbi–Viterbi) + C++17 tracking loops via pybind11; CMA → DD-LMS equaliser	QPSK end-to-end now; rest G2–G3
Blind coding ID (R-05/06, D1/D2)	Interleaver + code + scrambler identified with little data, incl. post-FEC scramblers	GF(2) linear algebra (bit-packed rank; NumPy now, C++ kernel planned), Berlekamp–Massey, Galois-field FFT for RS basis; soft rank profiles	Block interleaver + conv K3–9 + scrambler blind now; RS/LDPC G3
FEC decode (R-09)	Viterbi, RS, LDPC, concatenated — within 0.2 dB of reference implementations	C++17 kernels (soft Viterbi K3–9 punctured; RS BM/Chien/Forney dual-basis; normalised min-sum LDPC) + proven NumPy fallback (0.15 Mbit/s, 0 errors at 6 dB)	NumPy path works; C++ blocked on 64-bit toolchain (WP05)
Structure / Bit-Lens (R-10, D3)	Frames, sync, fields, CRC solved in GF(2), HEADER/PAYLOAD labels	FFT autocorrelation, GF(2) GCD/inverse CRC solving; ported from ui/src/lib/mock.ts	Working
Orchestrator (R-12)	One click → automatic analysis; anytime deep search	Python best-first hypothesis-tree search with memoisation, calibrated pruning (< 10^{-3} wrong-branch prune), budgets 8/64 attempts	v0 works blind (2–13 s); full G2
Layered proof (C1)	Proves the answer layer by layer; bounds false certificates ≤ 10^{-4}	NumPy re-synthesis + SciPy statistics (TOST equivalence tests, binomial parity tests) against a measured noise reference; near-miss battery	Working on Gen v0; calibration G2
Decode Certificate + verify (C1)	Signed, portable proof any third party can re-check in seconds	cryptography (Ed25519), canonical JSON, OS keyring; CLI sanjaya verify reuses the same proof code	Working; re-verify ≈ 1.7 s
Data sufficiency (C2)	“Is this enough data?” → predictive answer + collection tasking note	Tempered log-evidence ledgers, Wald bound, random-walk simulation for P50/P90	G4 (WP10)
Stitching + Sensor Twin (C3/C4)	Pools short captures from different sensors/locations into one decision	Weighted-least-squares network clock solution, GPS timestamps, log-likelihood pooling, per-capture proof	G4 (WP11); KiwiSDR validation
Recipe Compiler (C5)	Hands sensors a ready-to-run receiver built from the certified recipe	GNU Radio 3.10 .grc (stock blocks + embedded Python blocks); bit-exact release gate; recipe-matched detector (+15 dB for a 32-symbol sync)	G4 (WP12)
API layer	Engine ↔ UI, job control, binary tiles	FastAPI + uvicorn on 127.0.0.1; Pydantic → OpenAPI → generated TypeScript types; WebSockets; token + Host/Origin auth	Working (verified in Chrome)
UI	The frozen 5-step analyst flow with waterfall, constellation, bit views	React 19 + three.js (R3F), Vite, self-hosted fonts; Playwright visual regression to lock the design	Working
Desktop shell	Offline per-user installer incl. offline WebView2; no admin rights	Tauri 2 + PyInstaller engine sidecar; pinned-by-hash wheelhouse; CycloneDX SBOM; signed installers	G1 spike → G4 installers
Quality system	Nothing ships untested; every dependency has a fallback	pytest (30 tests), scripts/check.py layering rule + import-linter, planned ≥ 24 CPU-h fuzzing, nightly kill-matrix, commit-reveal sealed test sets	Working and green
Data engine	Training and validation data with exact truth, no hardware needed	SANJAYA-Gen: deterministic, seeded, per-layer taps; Watterson HF channel (ITU-R F.1487); real public recordings (KiwiSDR, SatNOGS/SatDump)	Gen v0 working; v1 taps G1
Independent oracles	Truth we did not create	SatDump, dumphfdl, fldigi, AFF3CT, liquid-dsp, gr-fec, galois — test-only, never shipped	Ongoing

PART E

Comparison with every public repository for this PS

Strict, rubric-based, reproducible — and honest about its limits

Method. Source: _build/data.json, a snapshot of GitHub repositories matching PS 26147 taken **26 Sep 2026** (metadata: stars, commits, languages, file lists; plus each repository's README and topics). **Hygiene first:** 52 entries − 1 off-topic result (a 2020 malware-evasion collection that merely matched the search) − 2 duplicate re-uploads − 2 forks of the same project − 3 documentation-only repositories (no code) = **44 distinct, code-bearing projects**. Each was scored by a fixed rubric on README claims *corroborated by file evidence* (demodulation claimed + a demod source file present scores higher than the claim alone).

Axis (max)	What earns points
PS coverage (35)	Claimed and file-evidenced: demodulation (7), de-interleaving (7), convolutional/Viterbi (7), Reed-Solomon (5), LDPC (5), concatenated (1), sampling-frequency handling (1), bit-stream correlation / header-payload (2)
Verification + honesty (15)	Test suite present (7), honest-limits language (4), CI (2), installable (2)
Real-signal engagement (15)	SigMF support (4), named real signals — Meteor, SatDump, STANAG, HFDL, NAVTEX, fldigi, KiwiSDR, SatNOGS, sigidwiki (2 each, cap 8), SDR hardware mention (3)
Engineering (20)	Commit count (8), code volume (6), recency of last push (3), GUI present (3)
Differentiation (15)	Scrambler identification (4), any verification mechanism (4), data-sufficiency (3), multi-capture / multi-sensor (2), GNU Radio output (2)

What the field looks like (44 projects). The PS's literal chain is well-trodden; the PS's *complaints* are not addressed by anyone.

Capability	Repos (of 44)	Share	Reading
Demodulation claimed	32	73 %	The PS baseline; nearly all claim it, file evidence for fewer
De-interleaving claimed	32	73 %	Mostly catalogue-based or hard-rank methods
Convolutional / Viterbi	27	61 %	
Reed-Solomon	23	52 %	
LDPC	22	50 %	Claimed; rarely with blind parameter identification
Header / payload correlation	32	73 %	
GUI	43	98 %	The PS asks for it; everyone builds it
ML-based classification	35	80 %	Mostly uncalibrated CNNs
Any test suite	26	59 %	Few have real coverage
SigMF support	11	25 %	
Named real signals used	4	9 %	Meteor / STANAG / NAVTEX / sigidwiki appear once each across the whole field
Scrambler identification	4	9 %	SANJAYA additionally handles the post-FEC coset case
Verification mechanism	0	0 %	Keyword hits exist (57 %) but none describes a signed, third-party re-checkable certificate; the closest anywhere is a CRC gate
Data-sufficiency answer	7	16 %	Mentions only; no mechanism like C2
Multi-capture stitching	1	2 %	Keyword only, no mechanism

Leaderboard (top 15 of 44; the SANJAYA row is self-assessed on the same rubric and flagged as such).

#	Score	Repository	PS	Ver	Real	Eng	Diff	One-line basis
1	74.0	The-4Script/RadioFry	35	6	4	20	9	100 commits, 200 files, all 5 PS stages file-evidenced, SigMF, GUI, ML; no tests, no named real signals, no verification
2	71.0	Devansh-567/hackathon	33	13	4	8	13	Full chain + scrambler + SigMF + tests + honest limits; but a single commit (one dump)
3	70.1	Venkata-Manoj/RF-signal-analysis	31	13	3	17	6	51 commits, 116 files, tests, GUI, ML, honest heuristics; no SigMF, no real signals
4	69.2	yoganandasp15/SIH_2026_Signal-Analyser	33	8	10	10	8	The only other repo engaging named real signals (NAVTEX, STANAG, sigidwiki) + SigMF + scrambler; 2 commits, no honest-limits language
5	65.2	arachknight66/SIH26147	33	13	4	15	0	40 commits, tests, SigMF, full-chain claims; zero differentiation beyond the PS
6	63.8	karurravishankermohit-stack/-SignalX	32	9	0	16	7	146 files, tests, GUI; no real-signal engagement, no scrambler
7	60.7	SomeNobody21112/ICHNOVA	19	13	2	18	9	75 commits, mentions KiwiSDR; but thin PS claims with no file evidence for demod/interleaver

#	Score	Repository	PS	Ver	Real	Eng	Diff	One-line basis
8	58.1	WaghJagdish/PS147	33	9	4	8	4	Full-chain claims with file evidence + tests + SigMF; single commit; no differentiators
9	57.1	Gururaghavendra123/sihps2-2026	33	13	0	7	4	Full chain + tests + honest limits; the field's only CRC-gate "verification"
10	56.1	EDM-Fan/Nova-Signum	33	9	0	8	6	Full-chain claims + tests; single commit
11	54.6	Krezmitent/SigNull	33	6	4	12	0	40 commits, SigMF; no tests, no differentiation
12	53.9	arunkumarmeda27/SpectraSync	23	13	0	14	4	144 files, tests; modest PS coverage
13	52.3	ashcodes1511/SIH-2026-Chroma-Code	26	12	3	7	4	SDR + scrambler mentions; modest coverage
14	51.2	xarjunpatil/SIH26147-Automated-model	26	13	0	3	9	Almost no code (2 files) yet honest + differentiated docs
15	50.9	ShantanuSomwanshi/SIH-RF-Signal-Intel	31	2	3	7	8	Strong claim set incl. scrambler + SDR; no tests, 17 files
—	86*	SANJAYA (self-assessed)	33	13	12	13	15	Working blind decode + layered proof + signed certificate + re-verify (measured); 7 named real signals planned; sufficiency / stitching / compiler unique; *self-score bias flagged

Head-to-head: SANJAYA vs the eight strongest repos. “Working” = demonstrated in this repository (engine verified end to end, 30 pytest tests); “G2–G4” = dated gate in the 10-week plan.

Capability	SANJAYA	RadioFry	Devansh-567	Venkata-Manoj	yogananda-sp15	arachknight66	SignalX	ICHNOVA	WaghJagdish
PS chain (i)–(v)	Yes (QPSK working; 10 classes G2–G3)	Claimed + files	Claimed + files	Claimed + files	Claimed + files	Claimed + files	Claimed + files	Thin claims	Claimed + files
Blind interleaver method	Soft, joint, ≤ 3N rows (block working)	Hard rank (typ.)	Hard rank (typ.)	Heuristic	Hard rank (typ.)	Hard rank (typ.)	Heuristic	—	Hard rank (typ.)
Scrambler ID incl. post-FEC	Yes, incl. coset (working)	No	Pre-FEC	No	Yes (pre-FEC)	No	No	No	No
Proof beyond bit level	Layered proof, working	No	No	No	No	No	No	No	No
Third-party re-check	Signed certificate + verify (≈ 1.7 s, working)	No	No	No	No	No	No	No	No
“Is this enough data?”	C2: P50/P90 + tasking note (G4)	No	No	No	No	No	No	No	No
Multi-sensor stitching	C3/C4 (G4, ≥ 3 KiwiSDR sites)	No	No	No	No	No	No	Keyword only	No
Receiver output for sensors	C5: .grc, bit-exact gate (G4)	No	No	No	No	No	No	No	No
Real signals, independent truth	Meteor-M vs SatDump + HFDL / NAVTEX (G2–G3)	None named	None named	None named	STANAG / NAVTEX named, truth unclear	None named	None named	KiwiSDR mention	None named
Test suite	30 pytest + layering CI	None found	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Honest-limits statement	Yes (schema limits stated)	Yes	Yes	Yes	No	Yes	No	Yes	No

Strict verdicts — who is better, and why.

vs repository	What they do better than SANJAYA today	What SANJAYA does better	Verdict
RadioFry (74.0)	Raw momentum: 100 commits and 200 files of shipped code; broader modulation surface claimed today	Verification (none there), real-signal truth, sufficiency, stitching, compiler; test suite absent there	**SANJAYA on substance** — the PS rewards provable confidence, not commit volume; RadioFry wins on cadence
Devansh-567 (71.0)	Nothing measurable; strongest README hygiene in the field	Provenance: 1 commit vs a working engine with 30 tests, measured proof results, signed certificates	**SANJAYA** — a claim set without history cannot be trusted or checked
Venkata-Manoj (70.1)	Iteration cadence (51 commits) and a larger codebase	Every decisive axis: proof, verification, real signals, sufficiency; they self-describe as heuristics	**SANJAYA**
yoganandasp15 (69.2)	Real-data instinct: the only other project naming real signals and SigMF; also lists scrambler ID	Mechanisms vs mentions: working blind decode + proof; truth rules (independent decoders); honest limits; schedule	**SANJAYA** — the closest competitor in instinct, behind in engineering
arachknight66 (65.2)	Cleaner commit history (40 commits vs our 5)	A moat: five interlocking capabilities they show none of; proof; tests equal, real signals ahead	**SANJAYA**
SignalX (63.8)	Code volume (146 files)	Real-signal engagement is zero there; no scrambler, no verification, no sufficiency	**SANJAYA**
ICHNOVA (60.7)	Commit count (75); KiwiSDR mention	Claims are thin with no file evidence for the core PS stages; SANJAYA’s core works	**SANJAYA**
WaghJagdish/PS147 (58.1)	Nothing measurable	Same case as Devansh-567: full claims, single commit, no differentiation	**SANJAYA**
Gururaghavendra123 (57.1)	The field’s only verification-like gate (CRC check) and honest limits	Layered proof is strictly stronger than a CRC gate; sufficiency, stitching, compiler, real-signal truth	**SANJAYA**
ShantanuSomwanshi (50.9)	SDR hardware ambition	Working software now; no tests there; 17 files of code	**SANJAYA**

Where competitors are genuinely ahead (the strict part). Three things, stated plainly. **(1) Public build momentum:** RadioFry shows 100 commits / 200 files and several teams show 30–75 commits; this repository shows about 5 — judges who weight visible iteration will notice. **(2) Shipped breadth today:** several repos claim working QAM/FSK/8PSK demos now, while the SANJAYA engine’s runtime slice is deliberately QPSK-only (schema 0.1) with the other classes dated at G2–G3. **(3) Star traction:** one docs-only repository has 30 stars; ours has none — irrelevant to capability, visible to skim-readers. None of these closes the structural gap: **0 of 44** repositories claim — let alone demonstrate — layered verification, a re-checkable signed certificate, a data-sufficiency mechanism, multi-sensor stitching with a clock model, or a bit-exact receiver compiler. Those five are interlocking (the compiler consumes certified recipes; stitching needs the Twin and the sufficiency maths; the proof arbitrates both), which is why copying any one of them does not reproduce the system.

Limitations of this comparison (read before quoting it). Scores are computed from README claims, topics and file lists as snapshotted on 26 Sep 2026 — not from running anyone's code, not a line-by-line audit, and not a benchmark. Repositories evolve daily; a repo behind today can lead next week. File-path evidence is shallow. The rubric's weights are judgment calls and are printed above precisely so they can be challenged. The SANJAYA row is self-assessed on the same rubric and should be read with that bias in mind. Earlier project documents ranked competitors manually (a different method) and named different leaders; the disagreement between the two methods is itself a reminder that these rankings are indicative, not definitive.

Bottom line. Against every public repository found for PS 26147, SANJAYA is the only project that already runs the PS chain blind on real files and then **proves** its answer with a signed certificate anyone can re-verify (≈ 1.7 s, measured) — and the only one even planning data-sufficiency, multi-sensor stitching, and a bit-exact receiver compiler. Competitors lead on none of the PS's stated pain points; they lead on commit momentum and claimed breadth, which we close at gates G2–G3 (25 Oct / 8 Nov). Strictly: **on the criteria the problem statement rewards — accuracy and confidence of interpretation — SANJAYA is ahead of all 44; on raw public velocity, one team (RadioFry) is ahead of us today.**

Sources: PRD.md v2.4 (§3 KPIs, §4 USPs, §6 architecture, §11 programme, §16 measured evidence) · HANDOFF.md 27 Sep 2026 (build state) · _build/data.json (52-entry snapshot, 26 Sep 2026) · _build/feasibility.py results. Competitor characterisations are README and metadata claims at snapshot time. Generated as a read-only analysis: no repository file was modified.